

Response to Reviewer

Latency Optimization of Cascaded Voice Dialogue Systems with a Pipeline-Parallel Streaming Architecture

Revision date: August 22, 2026

Dear Editor and Reviewer,

Thank you for identifying the principal evidence gaps in the original submission. We added concatenated human-read speech, a unified server-side first-playable-PCM timeline, distributional and repeated measurements, a configured LocalAgreement-2-style comparator, and instrumentation for commitment drift, tokenizer seams, and downstream responses. We also completed a final implementation audit and a deterministic CPU-only reanalysis of the locked records. No ASR, LLM, TTS, CUDA, or new data-generation experiment was run for this final correction.

The audit produced five scope corrections now made explicit throughout the paper. First, the historical ASR duration counter accumulated admitted speech and was not decremented after segment removal, so the 2.0-s condition was a latched startup gate rather than a recurring current-queue-duration threshold. Second, accumulated turns share source dialogues, so primary synthetic uncertainty now uses dialogue-cluster resampling. Third, run-log review confirmed that seven of the 505 candidate executions were contaminated by concurrent external programs, so all inferential analyses use the locked 498-turn valid set and retain the excluded records only in the audit ledger. Fourth, the first-audio comparison uses System B’s first sentence versus System A’s capped full response and is therefore a server-side whole-system policy comparison. Fifth, in that run ASR and CosyVoice shared GPU0 while Qwen2 used GPU1. We report the measured historical implementation and do not infer the performance of an unmeasured corrected trigger.

Comment 1: Real-recording robustness

Reviewer comment. *The main benchmark is synthesized from dialogue text, and the non-streaming ASR consequently has near-zero error. Please validate the system on real recordings with speaker, accent, noise, speaking-rate, and endpoint variation; these conditions directly affect VAD, timestamp stability, and transcript commitment.*

Response. We added long-form samples constructed from public human-read recordings rather than TTS. LibriSpeech test-clean utterances are concatenated within chapter, and AISHELL-1 utterances are concatenated in speaker order. Uniformly sampled pauses of 0.2–1.0 s introduce endpoint variation. Each language contains 75 clean samples, and six 30-sample conditions add SNR 20/15/10 dB noise, speed 0.9/1.1, or 15-dB babble. A five-sample listening check confirmed intelligibility and alignment while finding some audible seams; the paper therefore calls these data “concatenated human-read speech,” not spontaneous dialogue.

On clean Extra Long speech, historical-policy post-feed residual TTFT falls by 67.5% on LibriSpeech and 65.7% on AISHELL-1. Recognition quality is not assumed to be preserved: clean mean-utterance WER changes from 2.97% to 5.65% for English and CER from 10.77% to 11.80% for Chinese. No noninferiority margin, task-success criterion, user study, or usability claim is attached to these latency results. Babble remains a failure boundary: English System B is slower in the overall summary, empty outputs occur for 12/30 English and 5/30 Chinese samples, and one

queue requires about 21 s to drain.

Changes in the revised manuscript. The human-read construction and augmentation protocol appears in the data section, and the corresponding latency/error results are in Table VI and the concatenated-human-read subsection. The limitations now state that these are controlled concatenations measured with the historical latched policy rather than natural conversation.

Comment 2: Time to first audible output

Reviewer comment. *TTFT is measured from the end of speech to the first LLM token and excludes network and TTS latency. Since the paper discusses voice-response latency, please also report time to first audible output, including endpointing, communication, and initial TTS generation.*

Response. We now distinguish legacy post-feed residual TTFT from server-side time to first playable PCM. The unified run places physical speech end, feed end, input close, first token, policy-defined text readiness, TTS request, and the first accumulation of at least 1324 bytes (approximately 30 ms of 22.05-kHz mono 16-bit PCM) on one `perf_counter_ns` clock. All six components close exactly for every record. The endpoint excludes client transport, jitter buffering, audio-device scheduling, and acoustic playback, so we no longer call it user-perceived audibility.

The comparison is deliberately reported as a whole-system policy bundle. System B sends its first completed sentence to TTS, whereas System A sends its capped full response. Across 50 samples per mode, mean TTS text length is 17.74 versus 204.10 characters, 25/50 System B outputs are exactly one of two fixed Chinese greeting strings versus 0/50 for System A, and max-token stops occur in 43/50 versus 41/50. For English-input System B records, 19/25 TTS strings contain no Latin letters; Table VIII therefore labels this column “Input language.” Under these unequal policies, median server-side time to first playable PCM is 3.11 versus 22.27 s. We do not attribute the complete difference to ASR scheduling or incremental prefill and do not treat the TTS component as a matched-text comparison.

The prototype uses in-process ASR/LLM queues and localhost TTS. ASR and CosyVoice share GPU0, while Qwen2 uses GPU1. The 133.0-ms feed-end-to-close interval includes about 132.68 ms before explicit flush, 0.21 ms for flush, and 0.12 ms after flush; it is not flush-compute time.

Changes in the revised manuscript. Separate endpoint definitions now appear in the Introduction and evaluation-metrics section. Table VIII and its surrounding text disclose the response policy, TTS-text lengths, greeting and cap counts, input/output-language boundary, GPU placement, and absence of client playback.

Comment 3: Variability, percentiles, and repeated measurements

Reviewer comment. *Tables III–V mainly report mean values. Add standard deviations or percentile latency, especially P95/P99, and repeat measurements to show whether the approximately 1.1 s plateau remains stable under GPU contention and segment-trigger jitter.*

Response. Tables III–V report sample standard deviations and tail percentiles. Fig. 6 uses 12 equal-frequency duration bins with observed P5–P95 bands. We replaced a universal constant-

bound claim with a valid-set weaker-growth observation: System B P99 grows $1.32\times$ from Long to Extra Long, whereas System A grows $4.96\times$. Short and Medium regressions and the separately measured babble backlog remain visible.

The 1,132-turn primary set comes from 100 source dialogues, and the locked three-arm analysis set contains 498 turns from 99 dialogues. Synthetic-latency uncertainty now uses 10,000 dialogue-cluster bootstrap resamples; dialogue-level Wilcoxon tests operate on per-dialogue mean differences. The original-platform reduction is 74.34% with cluster 95% CI [72.62%, 75.76%]. The seven other candidate executions are not treated as system outcomes because their run logs confirmed contamination by concurrently executing external programs; they remain documented in the exclusion ledger.

We also repeated 50 Very Long turns from 40 dialogues three times on the second platform. System B round means were 1434.0, 1482.1, and 1454.9 ms; its per-sample CV had mean 5.19%, median 4.05%, P90 10.73%, and maximum 18.96%. No GPU-contention experiment was run. “Exclusive” now means no unrelated GPU job, not absence of ASR/TTS co-residency. The archived main runs also used fixed arm order, which is disclosed as a limitation for small contrasts.

Changes in the revised manuscript. Tables III–V and Fig. 6 now name the legacy endpoint and historical policy. Table IV documents the 505-candidate, seven-contaminated, 498-valid ledger and dialogue-cluster uncertainty. The statistical protocol and limitations document clustering, fixed order, tails, and the missing contention condition.

Comment 4: Stronger streaming comparator

Reviewer comment. *The comparison is limited to the authors’ non-streaming System A. Include a stronger streaming baseline or alternative chunked-ASR/prefill implementation, with matched models and hardware, so the gain can be attributed to the proposed scheduling and cache design.*

Response. We added a project-internal configured LocalAgreement-2-style implementation based on Liu et al. and the Whisper-Streaming adaptation of Macháček et al. It shares Whisper weights, inference engine, audio segmentation, hardware, and the locked valid 498-turn/99-dialogue set with System B. It uses absolute-time commitment, punctuation-robust agreement, sentence-aware trimming, and a 15-s retained-buffer cap. Trigger policy is not matched: System B uses the historical latched startup gate, while the comparator follows its new-audio/agreement policy. We therefore no longer call the comparison fully matched or generalize to the LocalAgreement family.

At this operating point, configured LA-2-style mean post-feed residual TTFT is 2115.0 ms versus 1573.9 ms for System B. The 541.1-ms paired difference has dialogue-cluster 95% CI [485.66, 599.14] and a dialogue-level Holm-adjusted $p = 1.14 \times 10^{-17}$. Recognition errors differ across systems and languages, so this is not a quality-equivalence result.

The ablation was also reinterpreted. Across the locked valid 498 turns, Baseline minus Streaming-ASR-Only is 3332.13 ms [3064.38, 3572.61], whereas Streaming-ASR-Only minus Full-Streaming is only 15.50 ms [10.95, 19.66]. Duration-stratified incremental-prefill point contrasts remain non-monotonic (−23.01, 2.73, and 40.82 ms). These fixed-order arm contrasts associate most measured reduction with streaming ASR; they do not establish a large independent causal benefit from incremental prefill.

Changes in the revised manuscript. The comparator subsection and Table VII use “configured LA-2-style” terminology and disclose the trigger-policy difference. Table IV replaces causal “gain” labels with arm contrasts and reports dialogue-cluster uncertainty.

Comment 5: Corrections, tokenizer boundaries, and downstream meaning

Reviewer comment. *Please clarify how committed text, tokenizer boundaries, and the KV cache are handled when a partial ASR hypothesis changes. Report rollback or correction frequency and evaluate downstream response quality, because WER/CER alone does not show whether incremental prefilling preserves dialogue meaning.*

Response. We define separate internal and downstream state. Internal ASR hypotheses may change when retained audio is recognized again, whereas a committed downstream fragment is delivered once and never withdrawn. There is no KV-cache rollback. On 50 Very Long accumulated turns from 40 dialogues, 375 downstream commits produce zero rollback deliveries; 224 internal changes affect 224/425 committed segments and 49/50 turns.

Fragment-wise and one-shot token sequences differ in 25/50 turns, although both decode back to identical text. This establishes text reconstruction only, not token, KV-state, logit, or output-distribution identity. The response-level evaluation remains exploratory: BGE-M3 cosine is 0.8832, while one intent judge gives A 3.10/5 and B 3.04/5 with B–A CI $[-0.34, 0.22]$. Independent response sampling, cap termination, one unversioned service judge, and the absence of emission-time committed WER/CER, entity/number/negation analysis, task success, human ratings, or a noninferiority margin prevent any claim of semantic or usability equivalence.

Changes in the revised manuscript. The methods define the append-only contract and tokenizer seam, while the state-analysis and limitations sections explicitly state what was and was not established.

Additional corrections from the final audit

- Legacy TTFT is now named post-feed residual TTFT; physical-speech-end TTFT and server-side first-playable-PCM latency are defined separately.
- The 2.0-s value is documented as a latched cumulative-seen-duration startup gate. Existing results measure this historical behavior; no corrected-trigger performance is claimed.
- The synthetic benchmark is described as a purposive accumulated-turn stress set, with source dialogue as the inferential cluster.
- The 498-turn valid set is used consistently; seven run-log-confirmed externally contaminated executions are documented but excluded from inferential results.
- Incremental prefill is retained as a scheduling component but is not presented as the dominant proven acceleration source.

- Fig. 2 states that both systems use standard decode-time KV caching; the difference is prompt-arrival scheduling.
- Claims of a universal routing threshold, constant bound, universal noise robustness, quality equivalence, and validated usability were removed or bounded.
- Related work now distinguishes Simul-Whisper, Prompt Cache, SGLang/RadixAttention, Page-dAttention, FlashAttention, and StreamingLLM from the mechanism evaluated here.

We believe these revisions directly address the five concerns while making the measured implementation, statistical unit, external-contamination exclusion, output policy, and evidentiary limits explicit.

Sincerely,

Haihua Mo and Zhengyou Liang